

Topic 2.9 - Logarithmic Expressions

Guided Notes

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____

Class: _____

Date: _____

Why this topic matters

A logarithm answers an exponent question. Every valid logarithm has a positive argument, a positive base different from 1, and a real output. Exact evaluation relies on recognizing powers; approximation uses change of base or calculator functions while preserving domain restrictions.

Learning targets

- Evaluate logarithmic expressions by translating between logarithmic and exponential forms.
- Approximate common, natural, and arbitrary-base logarithms with technology.
- Interpret logarithms as exponents in mathematical, scientific, and contextual scales.

Language and structure

Term	Meaning in this topic
logarithm	The exponent to which a base must be raised to produce a given positive number.
common logarithm	A logarithm with base 10, written $\log x$.
natural logarithm	A logarithm with base e , written $\ln x$.
argument	The positive expression inside a logarithm.
change of base	The identity $\log_b x = \frac{\ln x}{\ln b} = \frac{\log x}{\log b}$.

Launch: make a claim before calculating

Launch investigation

Reasoning first

Without a calculator, place $\log_2 20$ between two consecutive integers and explain the reasoning in exponential language.

Concept 1: Logarithmic and exponential forms

Core idea

$\log_b a = c$ if and only if $b^c = a$, where $b > 0$, $b \neq 1$, and $a > 0$.

- The base stays the same when moving between forms.
- The logarithm output is an exponent.
- Use the exponential form to solve simple equations involving an unknown argument, exponent, or base.

Worked example - annotate the reasoning

Problem. Rewrite $\log_3 81 = 4$ in exponential form and evaluate $\log_{1/2} 8$.

Model reasoning. $\log_3 81 = 4$ means $3^4 = 81$. For $\log_{1/2} 8 = c$, solve $(1/2)^c = 8 = 2^3$, so $2^{-c} = 2^3$ and $c = -3$.

Check your understanding**Independent**

Solve $\log_b 125 = 3$ for the positive base b .

Concept 2: Exact values and restrictions**Core idea**

Exact logarithms come from known powers and roots; the argument must remain positive.

- $\log_b 1 = 0$ because $b^0 = 1$.
- $\log_b b = 1$ because $b^1 = b$.
- Fractional arguments can produce negative logarithms when the base is greater than 1.

Worked example - annotate the reasoning

Problem. Evaluate $\log_4(1/32)$ exactly.

Model reasoning. Since $4 = 2^2$ and $1/32 = 2^{-5}$, solve $2^{2c} = 2^{-5}$, giving $c = -5/2$.

Check your understanding

Independent

State the domain of $\log_5(2x - 7)$.

Concept 3: Approximation and scale meaning

Core idea

Use change of base when the calculator does not have the desired base.

- Round only at the end of a calculation.
- A difference of 1 on a base-10 logarithmic scale corresponds to a factor of 10 in the underlying quantity.
- Interpret the result as an exponent or scale level, not as a raw amount.

Worked example - annotate the reasoning

Problem. Approximate $\log_7 250$ to three decimal places.

Model reasoning. $\log_7 250 = \ln 250 / \ln 7 \approx 2.838$.

Check your understanding

Independent

The pH model is $\text{pH} = -\log[H^+]$. Compare hydrogen-ion concentrations for pH 4 and pH 6.

Synthesis and exit ticket

Before you leave

Use precise notation, show the invariant or inverse structure, and interpret each parameter in context. A correct number without a defensible method is incomplete.

Exit ticket 1**No calculator**Evaluate $\log_9 3$.

Exit ticket 2**No calculator**Rewrite $5^{-2} = 1/25$ in logarithmic form.

Exit ticket 3**No calculator**State the domain of $\ln(4 - x)$.
